

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. (ME- CAD/CAM and AMT) Examination

May 2018

**ME767/ME709.01 Advanced Engineering Optimization Techniques
(ELECTIVE-IV)**

Date: 05.05.2018, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. Question paper comprises of two sections. Section I and II must be attempted in *separate answer sheets*.
2. Make suitable *assumptions* and *draw* neat figures wherever required.
3. Use of *scientific calculator* is allowed.

SECTION – I**Q-1** Answer following multiple choice questions. Write only correct choice in answer sheet. **[07]**

- (i) A machine is to be stopped for at least 30 minutes, if it continuously runs for 2 hours. If 'x' is the number of machine hours to be planned in a 8 hours day, we have constraint as:
- (p) $x \leq 8$ (q) $x \leq 6$
 (r) $x \leq 6.5$ (s) None of the above
- (ii) Production of 'x' pens cost INR 100 and that of 'y' pencils cost INR 120. If company wishes to optimize the unit costs, the objective function would be:
- (p) Min. $100x + 120y$ (q) Min. $x/100 + y/120$
 (r) Max. $-100/x - 120/y$ (s) $x \leq 100; y \leq 120$
- (iii) If $f(x^*) \leq f(x^* \pm h)$ then function $f(x^*)$ is definitely:
- (p) Local minima (q) Local maxima
 (r) Global minima (s) Global maxima
- (iv) A local minima may be greater than local maxima but it should be _____ global maxima.
- (p) Less than (q) Greater than
 (r) Equal to (s) Less than or equal to
- (v) The $\sin(x)$ curve shows:
- (p) Local minima at $x = \pi$ (q) Local maxima at $x = \pi$
 (r) Local as well as global maxima at $x = \pi/2$ (s) All of the above
- (vi) For a function $f(x) = 4x^5 - 25x^4 + 40x^3 + 10$, which of the following point will not satisfy the necessary condition for optimization:
- (p) $x = 0$ (q) $x = 1$
 (r) $x = 2$ (s) $x = 3$
- (vii) For a function $f(x) = 4x^5 - 25x^4 + 40x^3 + 10$, point of inflection is:
- (p) $x = 0$ (q) $x = 1$
 (r) $x = 2$ (s) $x = 3$

Q-2 Minimize following function within the interval (5, 9) with 10% accuracy using Interval Halving method. **[07]**

$$f(r) = \sin^{-1}\left(\frac{1}{r^2}\right) + \left(\frac{1}{r^2} - 10\right)\sin(r) + 15$$

OR**Q-2** Determine the minimum of the function $f = x^5 - 5x^3 - 20x + 5$ in the interval (0, 5) to within 10% of the exact value, using Dichotomous search method. Take $\epsilon = 0.001$. **[07]****Q-3** Answer the following (Attempt any three) **[21]**

- (a) Using Golden section search method, minimize the following function in the interval (1, 3). Perform 4 iterations.

$$f(x) = \frac{7}{2}x^2 + \frac{15}{x^3} \quad \text{on interval } 1 \leq x \leq 3$$

- (b) Briefly explain the working principal of Fibonacci search method with an illustration.
- (c) Minimize the following function using Newton method with initial solution point at $\lambda = 1$. Consider convergence criteria as 0.001 for value of solution point.

$$f = \frac{\lambda^3}{60} + \frac{\lambda^2}{20} + \frac{\lambda}{10} + \frac{5}{\lambda} + \frac{5}{3\lambda^2}$$

- (d) Find the dimension of the largest rectangle that can be inscribed in a circle of radius 1 meter using method of direct substitution.

SECTION – II

Q-4 Answer following multiple choice questions. Write only correct choices in answer sheet. **[05]**

- (i) In a function of two variables, Hessian matrix is neither positive nor negative definite, at a point. Then this point indicates:
- | | |
|----------------------------|----------------------------|
| (p) Global extreme point | (q) Relative maximum point |
| (r) Relative minimum point | (s) Saddle point |
- (ii) Which of the following is not a method to solve multivariable optimization problem with equality constraints:
- | | |
|--------------------------|---------------------------|
| (p) Direct substitution | (q) Constrained variation |
| (r) Lagrange multipliers | (s) None of the above |
- (iii) Given that $f(x) = x^2 + y^2 + z^2$ subject to $x - y = 0$ and $x + y + z = 10$. If this problem is to be solved by direct substitution method, the function can be reduced to the extent of:
- | | |
|---------------------|-------------------|
| (p) One variable | (q) Two variables |
| (r) Three variables | (s) No variables |
- (iv) Which of the following numerical methods of optimization use gradient of the function:
- | | |
|---------------------------|-----------------------|
| (p) Pattern search method | (q) Univariate method |
| (r) Descent method | (s) None of the above |
- (v) In univariate method, for a probe length ε , and if $-S_i$ is the correct search direction for minimizing a function f , then:
- | | |
|-------------------|-------------------|
| (p) $f_i^+ < f_i$ | (q) $f_i^- < f_i$ |
| (r) $f_i^- = f_i$ | (s) $f_i^+ > f_i$ |

Q-5 Using Powell's Pattern search method, minimize the function $f(x, y) = 3x^2 + 4y^2 - 5xy - 8y$. **[09]**
Take initial solution point as (0, 0).

OR

Q-5 Maximize the function $f(x, y) = 2x^2 - 7y^2 + 12xy$ using Kuhn-Tucker conditions. The problem **[09]**
is subjected to the constraint $2x + 5y \leq 98$; where $x, y \geq 0$.

Q-6 Answer the following (Attempt any three) **[21]**

- (a) Find the minimum and maximum distance of a point (3,4,12) from the sphere $x^2 + y^2 + z^2 = 1$. Use Lagrange multiplier method.
- (b) A cold storage godown of floor area 22500 square feet and height 18 feet is to be constructed in the rectangular form with brick walls on the three sides and asbestos sheet shutter on fourth side. If heat loss per unit area of floor, brick wall, roof and shutter is in the ratio 1:2:4:5, find the dimensions of the storage that minimizes heat loss. Use method of constrained variation.
- (c) Solve the following problem using interior penalty function method.
Minimize $f = 2x_1^2 + 3x_2^2 + x_1 + x_2$; Subject to $5 - x_1 \leq 0$, and $-x_2 \leq 0$
- (d) Enlist the non-traditional techniques used to solve optimization problems. Explain genetic algorithm techniques in detail.
- (e) Explain the structure of simulated annealing process. Enlist its advantages and disadvantages.
